HPAI Heterogeneous Spatiotemporal GNN Canada

Comprehensive Project Progress, Modeling Benchmarks & Research Decision Record — August 2026

---

1. Executive Summary & Core Objective

This project implements an automated, reproducible machine learning pipeline using **heterogeneous spatiotemporal graph neural networks (GNNs)** to forecast one-month-ahead Highly Pathogenic Avian Influenza (HPAI) poultry outbreak risk across Canadian regions. The primary prediction target is binary: *Will at least one poultry outbreak occur in Census Subdivision (CSD) r during month t+1, using only data available up to month t?*

Key Project Specifications:

- **Spatial Unit:** Statistics Canada Census Subdivisions (CSDs) — 5,161 total CSD nodes (5,054 provincial CSDs used in modeling).
- **Temporal Coverage:** Monthly steps from 2022 to 2025 (2022–2024 for training/validation, 2025 strictly held out for testing).
- **Extreme Class Imbalance:** Outbreak positive rate is ~0.16% (79 positive CSD-months out of 55,594 test observations in 2025; ratio ~622:1).

2. Data Pipeline & Multi-Modal Feature Extraction

A multi-modal geospatial data ingestion and processing pipeline has been established, uniting agricultural census metrics, road infrastructure, satellite landcover, and climate covariates into a unified modeling table:

- **Outbreak Labels:** CFIA outbreak records spatially assigned to CSD polygons and aggregated to monthly binary flags.
- **Poultry Agricultural Proxies:** Broiler, turkey, and poultry farm counts from the Census of Agriculture (CCS level lookup).
- **GLW Poultry Density:** Global Livestock Watch 2020 chicken and 2015 duck density summarized per CSD (mean, max, 95th percentile).
- **Road Infrastructure:** National Road Network (NRN) total length, road density, and distance to major highways per CSD.
- **GEE Dynamic World Landcover:** Monthly Dynamic World water, flooded vegetation, crops, and trees surface area metrics.
- **ERA5-Land Climate Pipeline:** Monthly temperature, total precipitation, and soil moisture anomalies extracted per CSD.

3. Graph Engineering & Adjacency Topology

To capture spatial diffusion and environmental connectivity between geographic units, multiple graph topology variants were built:

- **Spatial Contiguity Graphs:** Rook (shared boundaries), Queen (shared corners), and Nearest-Neighbor Fallback (for island CSDs).
- **kNN-3 Spatial Graph:** Distance-based k-nearest neighbor network (k=3, distance threshold ≤ 100 km).
- **Snapshot-Specific Multi-Relation Graphs (RGCN):** Incorporates geographic distance, commercial poultry similarity (cosine similarity ≥ 0.70), and expanding-window water habitat similarity constructed strictly from historical observations up to month t.

4. Benchmark Modeling Performance (Strict 2025 Test Split)

All models were evaluated on the unchanged 2025 held-out test split (N = 55,594 CSD-month observations, N_pos = 79). Performance is measured primarily via **PR-AUC** and top-K precision/recall due to extreme label imbalance:

Model Variant	Graph / Model Architecture	PR-AUC	ROC-AUC	Prec@50	Rec@50	Status
kNN-3 GraphSAGE (3-Seed)	kNN-3 Spatial Graph (k=3, d≤100km)	**0.0791**	0.9168	**0.2000**	**0.1266**	Valid Spatial Baseline
Relational GNN (3-Seed)	Snapshot Time-Safe Multi-Relation Graph	0.0698	0.9126	0.1600	0.1013	Primary Valid RGCN
Strict Env + Lag LR	Tabular Linear Baseline (No Graph)	0.0654	0.9084	0.1200	0.0759	Linear Baseline
Strict MLP (No Graph)	Tabular Neural Network	0.0601	0.9048	0.1200	0.0759	Non-Graph Neural Net
Time-Persistence Baseline	Historical Outbreak Recurrence (2022-2024)	0.0405	0.7311	**0.2000**	0.1270	Recurrence Baseline
Superseded RGCN (Static)	Static 2022-2024 Water Habitat Graph	0.0838*	0.9174	0.2000	0.1266	SUPERSEDED (Leakage)

5. Critical Methodological Decisions & Audit Findings

- **Time-Safety & Covariate Leakage Mitigation:** Identified that static multi-year graph averaging introduced future landcover leakage into early training snapshots. Replaced static graph construction with a *snapshot-specific time-safe graph pipeline*, establishing strict temporal isolation.
- **Cluster-Bootstrap Uncertainty Analysis:** Conducted 1,000 cluster-bootstrap resamples on 2025 CSD clusters. The 95% confidence interval for Relational GNN vs. Logistic Regression is **[-0.0240, +0.0338]** (includes zero). Formally designated RGCN performance as *comparable, not proven significantly superior* over linear baselines on static snapshot graphs.
- **Regional Performance Disparity:** RGCN achieved strong performance in British Columbia (PR-AUC 0.1750), moderate performance in the Prairies (0.0275), and lower performance in Quebec/Ontario (0.0096) and Atlantic Canada (0.0065).

6. Current Work & Next-Generation Architecture Roadmap

- **Wild Bird Reservoir Integration:** Incorporating GBIF and eBird wild waterfowl activity proxies to capture bird presence and seasonal migration pathways.
- **True Spatiotemporal Directed Dynamic GNN:** Transitioning from monthly disconnected snapshots to a unified dynamic spatiotemporal graph where each node is a (CSD, month) tuple, linked by forward directed temporal edges *(t → t+1)* to model infection memory and spatial wave propagation strictly leakage-safe.